

Remote Computing for Physicists

SSH, file transfer, and running computations on other computers



Today's goal

- Connect to another computer from your laptop.
- Run commands and computations on that computer.
- Transfer files in both directions.
- Keep long-running computations alive after disconnecting.
- See how the same ideas extend to lab servers and HPC clusters.

Your laptop does not have to do the computation.

Why should a physicist care?

- Experimental-control computers
- Lab workstations
- Departmental servers
- University computing clusters
- Remote instruments and collaborators' machines

CONNECT → TRANSFER → COMPUTE → RETRIEVE

Today's remote computer

A Raspberry Pi 4 running Raspberry Pi OS Desktop



Just enough networking

- IP address — identifies a device on a network.
- Hostname — a human-friendly name for a computer.
- Port — identifies a network service on that computer.
- Client — initiates a connection.
- Server — accepts the connection.

Laptop → raspberrypi-onboard.local → SSH (usually port 22)

Can I reach the Raspberry Pi?

LIVE DEMO

```
ping raspberrypi-onboard.local  
  
# or, if you know its IP address  
ping 192.168.1.50
```

- This tests whether your laptop can reach the Pi over the network.
- Then compare what is visible on your laptop and on the Pi.

SSH — Secure Shell

LIVE DEMO

```
ssh student@physics-pi.local
```

```
hostname
```

```
whoami
```

```
pwd
```

```
ls
```

When I type a command now, which computer executes it?

Make the remote machine visibly change

LIVE DEMO

```
ssh student@<rpi-ip>  
cd ~/orientation  
mkdir alice  
mkdir bob  
mkdir charlie
```

Pi File Manager
~/orientation

- Keep the Pi's file manager open on the projector.
- Watch the folders appear as students create them remotely.

SCP — send a file

LIVE DEMO

```
# Run this on your laptop  
scp simulation.py student@physics-pi.local:~/orientation/
```

Laptop

simulation.py

scp

Raspberry Pi

simulation.py appears

Run the computation remotely

LIVE DEMO

```
ssh student@<rpi-ip>  
  cd ~/orientation  
python3 simulation.py  
ls
```

- The Python interpreter is on the Pi.
- The Pi's CPU and memory perform the calculation.
- The output file is created on the Pi.

SCP — bring the result back

LIVE DEMO

```
# Run this on your laptop  
scp student@<rpi-ip>:~/orientation/results.csv .
```

Raspberry Pi
results.csv

scp

Laptop
results.csv

For whole projects: rsync

LIVE DEMO

```
rsync -avP project/ student@physics-pi.local:~/project/
```

- Useful for directories and repeated transfers.
- Run it again after changing one file: only the differences need to be transferred.
- -a: archive -v: verbose -P: progress/partial files

A common research problem

What if my calculation takes six hours?

Do I need to keep my laptop awake and SSH connected?

tmux — keep the remote session alive

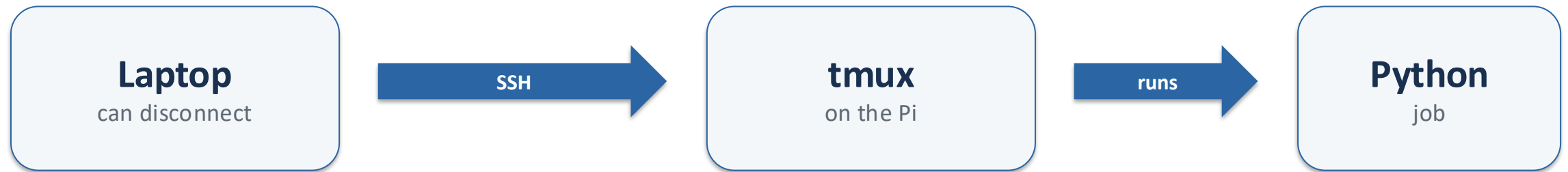
LIVE DEMO

```
tmux new -s physics
python3 long_calculation.py

# detach: Ctrl-b, then d

# later
ssh student@physics-pi.local
tmux attach -t physics
```

What survives when SSH disconnects?



The network connection can disappear while the computation continues on the Raspberry Pi.

SSH keys and ~/.ssh/config

Useful quality-of-life tools

```
ssh-keygen  
ssh-copy-id student@physics-  
pi.local
```

```
Host physics-pi  
HostName physics-pi.local  
User student
```

```
# then simply:  
ssh physics-pi
```

- Keep your private key private.
- The public key may be installed on machines you are authorized to access.

Bonus: remote Jupyter through SSH

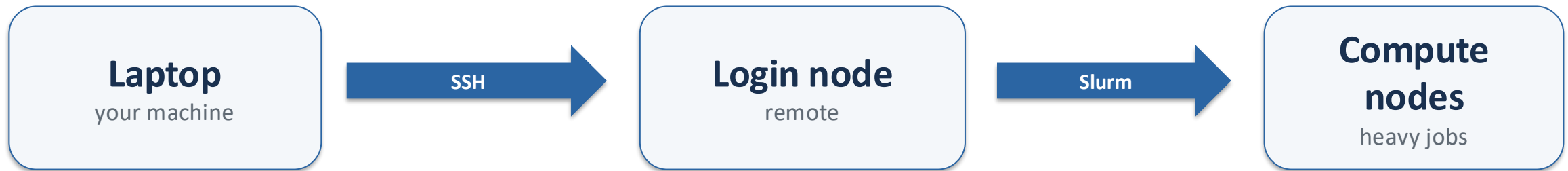
Only if time permits

```
# Jupyter is running on the Pi at port 8888  
ssh -L 8888:localhost:8888 student@<rpi-ip>
```

```
# On your laptop, open:  
localhost:8888
```

- This is SSH port forwarding (tunneling).
- A browser on your laptop can reach a service running on the remote computer.

From Raspberry Pi to a research cluster



- The machines get larger; the basic mental model remains.
- On HPC systems, submit heavy computation through the scheduler rather than running it on the login node.

Remote-computing survival card

```
# Connect  
ssh user@computer
```

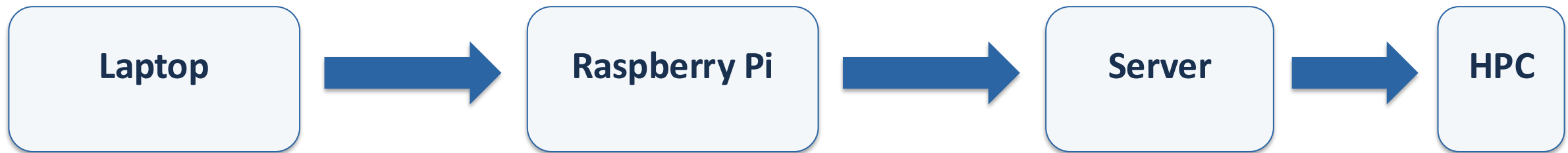
```
# Send / retrieve  
scp file user@computer:~/  
scp user@computer:~/file .
```

```
# Synchronize  
rsync -avP project/ user@computer:~/project/
```

```
# Persistent work  
tmux new -s work  
tmux attach -t work
```

One idea to remember

The computer in front of you does not have to be the computer doing your work.



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